Roll No.: DL.AI.UHAID24140

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech. Mid-Term Examinations – October 2024

First Semester

Artificial Intelligence

23PHY104 Computational Mechanics 1

Duration: Two hours Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Apply numerical methods to develop computational models for mechanical systems and analyze their behavior.
CO02	Derive constitutive relations for mechanical systems in motion or at rest, including particles and rigid bodies and use these equations to solve real world problems.
CO03	Evaluate the results of computational simulations and use this information to make informed decisions about mechanical systems design and optimization
CO04	Use software tools for computational mechanics, including code for solving equations of motion and simulating mechanical systems

PART A
Choose the best answer

1. Which of the following best explains why a seatbelt is important? [1] [CO01][BTL 1]
- a) Newton's First Law
- b) Newton's Second Law
- c) Newton's Third Law
- d) Law of Universal Gravitation

2. Which of the following is true according to Newton's Second Law? [1] [CO01][BTL 1]
- a) Force is directly proportional to mass
- b) Force is directly proportional to acceleration
- c) Force is inversely proportional to acceleration
- d) Force is constant irrespective of mass and acceleration

3. A force of 10 N is applied to a mass of 2 kg. What is the acceleration? [1] [CO02][BTL 4]
- a) 20 m/s^2
- b) 5 m/s^2
- c) 10 m/s^2
- d) 2 m/s^2

4. If a car goes from 0 to 100 km/h in 10 seconds, its acceleration is? [1] [CO02][BTL 4]
- a) 10 m/s^2
- b) 27.78 m/s^2
- c) 2.78 m/s^2
- d) 0.278 m/s^2

5. Which of the following is a scalar quantity? [1] [CO01][BTL 1]
- a) Force
- b) Velocity
- c) Energy
- d) Acceleration

6. If a force does no work on an object, which of the following is true? [1] [CO01][BTL 1]
- a) The object doesn't move
- b) The force acts perpendicular to the direction of motion
- c) The object moves in the same direction as the force
- d) The force increases the object's speed

7. Which of the following conditions must be met for a body to be considered in pure rotation? [1] [CO01][BTL 1]
- a) Every point on the body moves in a straight line
- b) Every point on the body moves with the same velocity
- c) Every point on the body moves along a circular path centered on the axis of rotation
- d) The body experiences both linear and angular displacement

8. Which of the following is true according to Newton's Second Law? [1] [CO02][BTL 2]
- a) Force is directly proportional to mass
- b) Force is directly proportional to acceleration
- c) Force is inversely proportional to acceleration
- d) Force is constant irrespective of mass and acceleration

9. The unit of power is: [1] [CO01][BTL 1]
- a) Newton
- b) Joule
- c) Watt
- d) Pascal

10. In rigid body kinematics, what is the angular velocity of a point on a rotating object if the radius is doubled and the linear velocity remains constant? [1] [CO02][BTL 4]
- a) Doubles
- b) Halves
- c) Stays the same
- d) Becomes zero

PART B
Answer all the questions.

1. How is position defined in mechanics, and why is a reference point essential? [2][CO01][BTL 1]
2. Explain the concept of velocity. How is it calculated from position? [2] [CO01][BTL 4]
3. Summarize Newton's Second Law of Motion and give an example of its application? [2] [CO01][BTL 2]
4. Describe the relationship between work and energy. How is work measured? [2] [CO01][BTL 2]
5. Define potential energy and describe the factors that affect it? [2] [CO01][BTL 1]
6. How is kinetic energy related to velocity, and what happens to kinetic energy if the velocity is doubled? [2] [CO02][BTL 1]

7. What is the significance of studying rigid body motion in engineering? [2] [CO02][BTL 4]
8. What conditions must be satisfied for an object to be in static equilibrium? [2] [CO01][BTL 1]
9. What information can be obtained from a free body diagram, and how is it constructed? [2] [CO02][BTL 1]
10. Explain angular displacement and how it is measured in rotational motion? [2] [CO02][BTL 4]

PART C
Answer all questions

1. Explain the different methods used to represent rigid body rotation, such as rotation matrices, Euler angles, axis-angle representation? [6][CO01][BTL 4]
2. Explain six basic commands in MATLAB along with their explanations? [6][CO03][BTL 4]

PART D
Answer all questions

1. What is the process for deriving the 2D rotation matrix, and what are the key steps involved? [8][CO02][BTL 1]

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CO	Marks	BTL	Marks
CO01	24	BTL 1	24
CO02	20	BTL 2	5
CO03	6	BTL 3	-
CO04	-	BTL 4	21